

QBITEL BRIDGE

AVIATION & AEROSPACE — DEPLOYMENT CHECKLIST

11-Phase Implementation Guide

- Phase 0: Pre-Engagement Safety Assessment
- Phase 1: Protocol Discovery and Analysis
- Phase 2: Ground Infrastructure Readiness
- Phase 3: Security Architecture Design
- Phase 4: ADS-B Authentication Deployment
- Phase 5: Data Link Protection (LDACS/ACARS/SATCOM)
- Phase 6: AI Spoofing Detection Activation
- Phase 7: ATC System Integration
- Phase 8: DO-326A Evidence Generation
- Phase 9: Validation and Testing
- Phase 10: Operational Handover
- Appendix: Performance SLAs, Rollback, Escalation

180K+

Daily Unauthenticated
ADS-B

36 Wks

Full Deployment
Timeline

99.999%

Availability SLA
Target

Zero

Aircraft Changes
Required

How to Use This Checklist

Priority	Badge	Meaning
REQUIRED	REQ	Must be completed before proceeding to next phase
RECOMMENDED	REC	Strongly recommended — omission requires documented justification
OPTIONAL	OPT	Optional enhancement — complete if applicable to deployment

PHASE
0

PRE-ENGAGEMENT SAFETY ASSESSMENT

Duration: 4 weeks | Owner: QBITEL Aviation Practice Lead + Customer Safety Team

Safety Team Briefing

- 1 ☐ **REQ** **SAFETY** Conduct QBITEL aviation security briefing for customer safety team
Include SMS coordinator, chief engineer, and airworthiness authority liaison
- 2 ☐ **REQ** **SAFETY** Review existing safety case and safety management system documentation
Identify existing security-relevant assumptions in safety case
- 3 ☐ **REQ** **SAFETY** Confirm QBITEL deployment has zero impact on existing safety cases
Ground-only: no impact. Airborne: ARINC 653 isolation assessment required
- 4 ☐ **REQ** **SAFETY** Document safety team acceptance sign-off for deployment scope
Formal acceptance required before any infrastructure work begins

DO-326A Gap Assessment

- 5 ☐ **REQ** **DO-326A** Initiate DO-326A Aircraft Security Log (ASL) with baseline security posture
ASL must capture security assumptions at deployment start for traceability
- 6 ☐ **REQ** **DO-326A** Conduct DO-326A gap analysis against existing security documentation
Compare current state against DO-326A requirements and identify gaps
- 7 ☐ **REQ** **DO-326A** Define security requirements derived from threat analysis (Security Target Analysis)
STA must cover ADS-B, ACARS, SATCOM, ATC network, and any airborne elements
- 8 ☐ **REQ** **SCOPE** Confirm scope: ground-only vs. ground + airborne partition deployment
Airborne deployment triggers DO-178C scope and STC/CS-STAN assessment

Airspace Authority Notification

- 9 ☐ **REQ** **REGULATORY** Notify national aviation authority (FAA/EASA/CAA) of planned security deployment
Use AC 119-1 (FAA) or AMC 20-42 (EASA) notification procedure

☐ 10 **REQ** **REGULATORY** Notify ANSP of planned ground receiver infrastructure changes
ADS-B receiver modifications require ANSP coordination

☐ 11 **REQ** **REGULATORY** Confirm no NOTAM requirements for ground infrastructure deployment
Ground-only deployment typically does not require NOTAMs

☐ 12 **REQ** **REGULATORY** Initiate FAA Issue Paper or EASA CRI if airborne deployment is in scope
Early engagement with authority is critical for airborne certification timeline

Preliminary Technical Assessment

☐ 13 **REQ** **TECHNICAL** Conduct ADS-B receiver network survey — inventory all receiver sites
Document receiver location, hardware, firmware, and network connectivity

☐ 14 **REQ** **TECHNICAL** Measure ADS-B message volumes at representative receiver sites
Required for authentication node sizing and throughput planning

☐ 15 **REQ** **TECHNICAL** Assess existing ATC network topology for QBITEL integration points
Document ASTERIX feeds, SDPS connections, SWIM interfaces

**PHASE
1**
PROTOCOL DISCOVERY AND ANALYSIS

Duration: 4 weeks | Owner: QBITEL Technical Team + ANSP Engineering

ADS-B Receiver Analysis

☐ **REQ** **ADS-B** Deploy passive ADS-B receiver tap at representative sites (minimum 5)
16 *Record and analyze 72 hours of ADS-B traffic for baseline characterization*

☐ **REQ** **ADS-B** Characterize ADS-B message volume, message rate, and ICAO address distribution
17 *Required for AI model training dataset and authentication enrollment planning*

☐ **REC** **ADS-B** Identify existing MLAT receiver infrastructure and integration capability
18 *MLAT integration significantly enhances spoofing detection accuracy*

☐ **REQ** **ADS-B** Analyze ADS-B message format distribution (ADS-B OUT types 0-31)
19 *Ensure QBITEL authentication handles all message type formats in use*

☐ **REC** **ADS-B** Identify any non-standard or military ADS-B traffic in airspace
20 *Military and government aircraft may require special handling configuration*

ACARS and Data Link Analysis

☐ **REQ** **ACARS** Inventory all ACARS VHF ground stations in scope
21 *Document frequency, ground station hardware, and network connectivity*

☐ **REQ** **ACARS** Monitor and catalog ACARS message types and volumes for 72 hours
22 *Required for ACARS gateway sizing and authentication overhead assessment*

☐ **REC** **ACARS** Assess HF ACARS infrastructure if oceanic routes are in scope
23 *HF ACARS requires separate gateway configuration for low-bandwidth constraints*

☐ **REC** **SATCOM** Survey SATCOM ground earth stations (Iridium, Inmarsat) if in scope
24 *SATCOM security requires gateway deployment at each ground earth station*

ARINC 429 Bus Analysis (Airborne Deployments Only)

☐ **REQ** **AIRBORNE** Map ARINC 429 bus topology for target aircraft type(s)
25 *Required for ARINC 653 security partition I/O specification*

☐ **REQ** **AIRBORNE** Identify transponder ARINC 429 output bus for TAU installation
26 *TAU connects as passive tap on transponder output bus*

☐ **REQ** **AIRBORNE** Review IMA platform ARINC 653 partition allocation and available time budget
27 *Security partition time budget must not affect safety-critical partition guarantees*

**PHASE
2**
GROUND INFRASTRUCTURE READINESS

Duration: 4 weeks | Owner: QBITEL Infrastructure Team + Customer IT/Network Team

ATC Network Segmentation

REQ
NETWORK

Assess ATC network architecture for QBITEL node placement

Authentication nodes placed between receiver feeds and SDPS/display systems

28


REQ
NETWORK

Implement network segmentation separating authentication infrastructure from ATC safety ...

QBITEL nodes must be network-isolated from safety-critical ATC automation

29


REQ
NETWORK

Deploy N+1 redundant QBITEL authentication nodes at each primary ATC facility

Minimum 2 nodes per facility; 3 for major Area Control Centres

30


REQ
NETWORK

Establish out-of-band management network for QBITEL infrastructure

Management access must not share paths with operational data flows

31


REQ
NETWORK

Deploy quantum-safe VPN between ATC facilities (QBITEL Bridge QS-VPN)

PQC-secured VPN mesh for inter-facility coordination and QBITEL management

32

Hardware Security Module (HSM) Deployment

REQ
PKI

Procure FIPS 140-3 Level 3 HSMs for root key storage

Minimum 2 HSMs per deployment (primary + backup) in geographically separate facilities

33


REQ
PKI

Initialize HSMs and generate root key material under dual-control ceremony

Key ceremony must be documented for DO-326A ASL. Witnesses required.

34


REQ
PKI

Configure HSM for QBITEL authentication workloads — Falcon-512 and ML-DSA acceleration

Verify HSM performance meets QBITEL throughput requirements under load

35

Certificate Authority Infrastructure

REQ
PKI

Deploy QBITEL Aviation PKI root CA (offline HSM-backed)

Root CA must remain offline except during subordinate CA issuance ceremonies

36


REQ
PKI

Deploy online subordinate CAs for aircraft enrollment and receiver authentication

Separate subordinate CAs for aircraft identity and receiver infrastructure

37


REQ
PKI

Establish CRL (Certificate Revocation List) and OCSP distribution points

Emergency revocation must propagate within 60 seconds to all ground stations

38


REQ
PKI

Configure certificate lifecycle management for automatic renewal

Auto-renewal initiated 90 days before expiration; completed 30 days before

39

REC

PKI

Test PKI federation with any partner ANSP PKI infrastructure

40

Cross-certification required for international airspace coverage

**PHASE
3**
SECURITY ARCHITECTURE DESIGN

Duration: 4 weeks | Owner: QBITEL Security Architects + Customer Chief Engineer

Deployment Model Decision

- ☐ 41

REQ

ARCHITECTURE

Confirm final deployment scope: ground-only vs. ground + airborne

Document decision rationale in DO-326A ASL for regulatory record
- ☐ 42

REQ

ARCHITECTURE

Define authentication enrollment targets: which aircraft types and operators

Prioritize high-frequency operators and aircraft with significant spoofing risk
- ☐ 43

REQ

ARCHITECTURE

Define behavioral authentication thresholds for non-enrolled aircraft

False positive rate target, detection sensitivity, and alert escalation thresholds
- ☐ 44

REQ

ARCHITECTURE

Design ATC display integration for authentication status indicators

HMI design reviewed with controllers and human factors specialists

ARINC 653 Partition Design (Airborne Only)

- ☐ 45

REQ

AIRBORNE

Define ARINC 653 security partition time budget and memory allocation

Security partition time budget must not affect safety-critical partition guarantees
- ☐ 46

REQ

AIRBORNE

Design security partition interfaces — ARINC 429 in, ARINC 429 out

Interface specification reviewed by aircraft OEM avionics engineering
- ☐ 47

REQ

AIRBORNE

Define security partition health monitoring interface to IMA health management

Partition health status reported per ARINC 653 health monitoring API
- ☐ 48

REQ

AIRBORNE

Complete DAL-D safety assessment and coordinate with OEM safety team

Confirm DAL-D is appropriate given partition isolation characteristics

Bandwidth and Compression Configuration

- ☐ 49

REQ

PQC

Configure PQC algorithm selection per link type based on Phase 1 measurements

LDACS: Falcon-512 compressed. VHF ACARS: Falcon-512 session. HF: session-based MAC.
- ☐ 50

REQ

PQC

Set session key cache TTL based on operational patterns

Default: 24 hours or 1,000 messages. Adjust for high-frequency operators.
- ☐ 51

REQ

PQC

Validate compressed PQC sizes against link frame capacity for all link types

Verify no link type exceeds overhead targets from Section 4 specifications

PHASE
4
ADS-B AUTHENTICATION DEPLOYMENT

Duration: 6 weeks | Owner: QBITEL Field Engineering + ANSP Ground Operations

Cryptographic Binding Infrastructure

- | | | |
|---|---|--|
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 52 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Deploy QBITEL authentication nodes at all ADS-B receiver sites in scope</p> <p><i>Authentication nodes installed in-line between receiver feed and network</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 53 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Configure cryptographic MAC verification with PKI integration at each node</p> <p><i>Verify node can validate MACs for enrolled aircraft in less than 50ms</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 54 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Deploy transponder-adjacent authentication units (TAUs) on first enrolled aircraft</p> <p><i>TAU installation during scheduled maintenance — estimated 4-8 hours per aircraft</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 55 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Enroll first aircraft operator cohort in aviation PKI</p> <p><i>Issue enrollment certificates, configure aircraft ICAO address binding</i></p> |

Message Authentication Code (MAC) Integration

- | | | |
|---|---|---|
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 56 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Verify MAC computation and verification for enrolled aircraft at each receiver site</p> <p><i>Perform end-to-end MAC test flights with known enrolled aircraft</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 57 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Configure authentication status pass-through to ASTERIX and SDPS feeds</p> <p><i>Authentication status bits embedded in ASTERIX records for ATC display</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 58 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Verify ATC display shows authentication status for enrolled aircraft</p> <p><i>Controller display shows verified/unverified/anomaly status correctly</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 59 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">ADS-B-AUTH</div> | <p>Validate graceful degradation — authentication failure does not block ATC data</p> <p><i>Simulate authentication node failure; confirm ADS-B data continues unaffected</i></p> |

PKI and Certificate Lifecycle Test

- | | | |
|---|--|---|
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 60 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">PKI</div> | <p>Test certificate revocation propagation: measure time from revoke to ground station awareness</p> <p><i>Revocation must propagate to all stations within 60 seconds target</i></p> |
| <div style="border: 1px solid #ccc; width: 20px; height: 20px; margin-bottom: 5px;"></div> 61 | <div style="background-color: #333; color: white; padding: 2px 5px; font-weight: bold;">REQ</div> <div style="background-color: #add8e6; padding: 2px 5px; font-weight: bold;">PKI</div> | <p>Test certificate renewal process: renew an enrolled aircraft certificate without interruption</p> <p><i>Renewal must complete without affecting authentication continuity</i></p> |

**PHASE
5**
DATA LINK PROTECTION — LDACS / ACARS / SATCOM

Duration: 4 weeks | Owner: QBITEL Data Link Team + Airline Operations / ANSP

LDACS PQC Configuration

☐ **REQ** **LDACS** Deploy QBITEL LDACS security gateway at ground station infrastructure
LDACS gateway intercepts, authenticates, and forwards data link traffic

☐ **REQ** **LDACS** Configure Falcon-512 compressed PQC for LDACS data sessions
Verify compressed signature fits within LDACS frame structure (600 bps minimum)

☐ **REQ** **LDACS** Test CPDLC authentication through LDACS security layer
Send test CPDLC clearances and verify authentication and integrity end-to-end

☐ **REQ** **LDACS** Validate LDACS session continuity across aircraft movement between LDACS cells
Session handover must maintain authentication without visible delay

ACARS Gateway Quantum Protection

☐ **REQ** **ACARS** Deploy QBITEL ACARS authentication gateway at all AOC ACARS uplink points
Gateway performs MAC authentication on all ACARS uplinks and downlinks

☐ **REQ** **ACARS** Configure VHF ACARS session-based PQC (Falcon-512 compressed)
Verify session establishment completes within 3 ACARS message exchange cycles

☐ **REC** **ACARS** Configure HF ACARS low-bandwidth authentication mode
Session-based MAC with 32-byte HMAC. Verify overhead under 12% at 1,200 bps.

☐ **REQ** **ACARS** Test ACARS authentication with representative message types (OOOI, M-PDU, weather)
Verify authentication does not exceed 0.8-second latency overhead on VHF

☐ **REQ** **ACARS** Enable non-repudiation logging for safety-critical ACARS messages
Maintenance write-ups and MEL deferrals logged with cryptographic proof of origin

SATCOM Session Security

☐ **REC** **SATCOM** Deploy QBITEL SATCOM security gateway at Iridium and Inmarsat ground earth stations
Gateway wraps SATCOM sessions with PQC-derived session keys

☐ **REC** **SATCOM** Configure adaptive PQC selection based on SATCOM link rate
Iridium (600 bps): session MAC. Inmarsat SB-S (10.5 kbps): full Falcon-512.

☐ **REC** **SATCOM** Test ACARS-over-SATCOM (AOS) authentication for oceanic routes
Verify authentication continuity during oceanic flight (Iridium and Inmarsat)

**PHASE
6**
AI SPOOFING DETECTION ACTIVATION

Duration: 6 weeks | Owner: QBITEL AI/ML Team + ANSP Operations

AI Multilateration Cross-Validation

- | | | | |
|-----------------------------|------------|------------------|---|
| ☐ 74 | REQ | AI-DETECT | Integrate QBITEL AI engine with MLAT receiver network feeds
<i>Minimum 4 receiver stations per coverage zone for MLAT accuracy</i> |
| ☐ 75 | REQ | AI-DETECT | Collect 4 weeks of baseline ADS-B+MLAT correlated data for AI model training
<i>Training dataset must include diverse aircraft types, weather, and traffic conditions</i> |
| ☐ 76 | REQ | AI-DETECT | Train AI behavioral models on local traffic patterns and aircraft performance data
<i>Model must be trained per aircraft type using locally representative training data</i> |
| ☐ 77 | REQ | AI-DETECT | Validate AI model performance against known legitimate anomalies
<i>Use historical data with known emergency maneuvers, weather deviations, equipment issues</i> |

Alert Threshold Configuration

- | | | | |
|-----------------------------|------------|------------------|--|
| ☐ 78 | REQ | AI-DETECT | Configure alert thresholds for position discontinuity detection
<i>Threshold based on aircraft performance envelope for each enrolled aircraft type</i> |
| ☐ 79 | REQ | AI-DETECT | Configure MLAT-vs-ADS-B position discrepancy alert thresholds
<i>Start conservative (high threshold) and tune based on false positive rate observations</i> |
| ☐ 80 | REQ | AI-DETECT | Configure velocity and altitude violation detection thresholds
<i>Velocity and rate-of-climb limits derived from aircraft type performance data</i> |
| ☐ 81 | REQ | AI-DETECT | Set alert escalation levels: Advisory / Warning / Alert for ATC display
<i>Advisory: anomaly detected. Warning: probable spoofing. Alert: confirmed attack.</i> |

False Positive Rate Validation

- | | | | |
|-----------------------------|------------|------------------|---|
| ☐ 82 | REQ | AI-DETECT | Operate AI detection in advisory-only mode for 2 weeks
<i>Record all detections; classify as true positive, false positive, or uncertain</i> |
| ☐ 83 | REQ | AI-DETECT | Measure false positive rate against target of less than 0.001%
<i>Adjust thresholds if false positive rate exceeds target before operational activation</i> |
| ☐ 84 | REQ | AI-DETECT | Conduct spoofing simulation exercises to validate detection rate
<i>Use QBITEL spoofing simulation tool in controlled, coordinated environment</i> |
| ☐ 85 | REQ | DO-326A | Document detection rate and false positive rate for DO-326A SDA
<i>Measured performance must be captured in Security Development Assurance documentation</i> |

PHASE
7
ATC SYSTEM INTEGRATION

Duration: 4 weeks | Owner: QBITEL Integration Team + ANSP Systems Engineering

Radar Correlation Integration

☐ **REQ** **ATC-INT** Integrate QBITEL radar correlation engine with SSR Mode S radar feeds
 86 *Ingest SSR data for real-time ADS-B vs. radar position correlation*

☐ **REQ** **ATC-INT** Configure radar-ADS-B discrepancy alert thresholds
 87 *Discrepancy threshold accounts for GPS vs. SSR accuracy differences*

☐ **REQ** **ATC-INT** Validate correlation engine performance with known radar-ADS-B offset data
 88 *Use historical data from aircraft with known GPS-SSR offsets for calibration*

Flight Data Processor Integration

☐ **REQ** **ATC-INT** Integrate authentication status with Flight Data Processor (FDP) track records
 89 *Authentication status appended to FDP track data for correlation with flight plan data*

☐ **REQ** **ATC-INT** Verify authentication status displayed on controller working positions
 90 *HMI verification with at least 3 controller working positions across different display types*

☐ **REC** **ATC-INT** Test ASTERIX output with authentication status bits to all downstream consumers
 91 *Verify all ASTERIX consumers (airline flight tracking, airport systems) receive status*

Safety Alert Integration

☐ **REQ** **ATC-INT** Integrate spoofing alerts with ATC safety net systems
 92 *Spoofing alerts coordinated with STCA, MSAW, and other safety net systems*

☐ **REQ** **ATC-INT** Verify spoofing alerts do not generate spurious STCA advisories
 93 *STCA must filter alerts for tracks flagged as spoofed/unverified*

☐ **REQ** **ATC-INT** Conduct integrated test with safety nets: confirm expected alert behavior
 94 *Simulated spoofing injection: verify correct alert hierarchy and suppression*

PHASE
8
DO-326A EVIDENCE GENERATION

Duration: 4 weeks | Owner: QBITEL Compliance Team + Customer Airworthiness Team

Security Target Analysis (STA)

- ☐

95

REQ

DO-326A

Complete Security Target Analysis covering all threat vectors in scope

STA must cover ADS-B spoofing, data link attacks, network intrusion, insider threats
- ☐

96

REQ

DO-326A

Document threat likelihood and impact assessments for each threat vector

Likelihood and impact based on operational environment, not generic assessments
- ☐

97

REQ

DO-326A

Derive security requirements from STA threat analysis

Each threat must have at least one security requirement that mitigates it

Security Development Assurance (SDA)

- ☐

98

REQ

DO-326A

Complete Security Development Analysis demonstrating requirements are implemented

SDA must trace each security requirement to design decisions and verification evidence
- ☐

99

REC

DO-326A

Include penetration test results in SDA as verification evidence

Third-party penetration test recommended for ATC-critical deployments
- ☐

100

REQ

DO-326A

Include ADS-B spoofing simulation results in SDA

Detection rate and false positive rate measurements from Phase 6

Aircraft Security Log (ASL)

- ☐

101

REQ

DO-326A

Update ASL with all security-relevant decisions made during deployment

ASL captures design decisions, assumptions, and accepted residual risks
- ☐

102

REQ

DO-326A

Obtain airworthiness authority acceptance of ASL (if airborne deployment)

Ground-only: ASL submitted to authority as notification. Airborne: formal acceptance required.

DO-178C Evidence (Airborne Partition Only)

- ☐

103

REQ

DO-178C

Deliver complete DO-178C Level D evidence package to OEM/STC applicant

Software Plans (4), Development artifacts (4), Verification artifacts (4)
- ☐

104

REQ

DO-178C

Coordinate DO-178C evidence review with FAA DER or EASA DOA

Designated Engineering Representative or Design Organization Approval review

PHASE
9
VALIDATION AND TESTING

Duration: 4 weeks | Owner: QBITEL QA Team + ANSP Safety Review Board

Spoofing Simulation Exercises

☐ **REQ** **VALIDATION** Conduct controlled ADS-B ghost aircraft injection test at non-operational receiver
 105 *Coordinated exercise with ANSP, using QBITEL simulation tool on isolated receiver*

☐ **REQ** **VALIDATION** Conduct ADS-B position falsification test for enrolled aircraft
 106 *Inject false position data; verify detection within 30-second target*

☐ **REQ** **VALIDATION** Conduct ACARS message forgery simulation at AOC test environment
 107 *Inject forged ACARS message; verify real-time authentication rejection*

☐ **REC** **VALIDATION** Test network intrusion simulation against ATC quantum-safe VPN
 108 *Third-party penetration test of VPN and SWIM API security layer*

False Positive Rate Final Validation

☐ **REQ** **VALIDATION** Measure operational false positive rate over 30-day monitoring period
 109 *Target: less than 0.001%. Document and investigate all false positives.*

☐ **REQ** **VALIDATION** Confirm false positive rate does not adversely affect controller workload
 110 *Controller workload assessment with operational air traffic controllers*

Safety Review Board

☐ **REQ** **SAFETY** Present DO-326A evidence package to internal Safety Review Board
 111 *Board includes safety team, airworthiness representative, and operations management*

☐ **REQ** **SAFETY** Obtain Safety Review Board acceptance before operational activation
 112 *Board acceptance documented as formal record in DO-326A ASL*

☐ **REQ** **REGULATORY** Submit DO-326A compliance package to national aviation authority
 113 *FAA: Issue Paper resolution. EASA: CRI closure. National CAA: equivalent process.*

**PHASE
10**
OPERATIONAL HANDOVER

Duration: 4 weeks | Owner: QBITEL Customer Success + ANSP Operations Management

ANS Operator Training

114 ☐ **REQ** **TRAINING** Complete QBITEL Bridge administrator training for ANS technical staff
System administration, monitoring, alert management, and key lifecycle management

115 ☐ **REQ** **TRAINING** Complete ATC controller familiarization training
Authentication status indicators, alert response procedures, reporting protocols

116 ☐ **REQ** **TRAINING** Complete SOC operator training for QBITEL monitoring workflows
Alert triage, incident classification, escalation procedures, regulatory reporting

117 ☐ **REQ** **TRAINING** Deliver operator documentation: system manual, quick reference cards, procedures
All documentation reviewed and approved by ANSP operations management

Safety Case Approval

118 ☐ **REQ** **SAFETY** Obtain formal safety case acceptance from national aviation authority
Ground-only: authority notification with no-objection response. Airborne: formal acceptance.

119 ☐ **REQ** **SAFETY** Update operational procedures and ATC controller instructions
Safety-case-approved procedures published and effective before operational activation

120 ☐ **REQ** **REGULATORY** Confirm no outstanding DO-326A evidence items or authority queries
All authority questions resolved before declaring Full Operational Status

Continuous Monitoring Activation

121 ☐ **REQ** **OPERATIONS** Activate QBITEL SOC monitoring for full operational coverage
24/7 monitoring with aviation-specific threat intelligence and alert triage

122 ☐ **REQ** **OPERATIONS** Configure automated regulatory incident reporting
Reporting templates compliant with FAA, EASA, and national CAA requirements

123 ☐ **REQ** **OPERATIONS** Establish monthly security posture review cadence with ANSP safety team
Monthly report includes detection statistics, false positive rate, and threat intelligence

124 ☐ **REQ** **DO-326A** Schedule annual DO-326A compliance review aligned with SMS audit cycle
Annual review updates ASL, reviews threat landscape, and updates STA if required

125 ☐ **REQ** **OPERATIONS** Declare Full Operational Status (FOS) with documented authority confirmation
FOS declaration recorded in DO-326A ASL and communicated to all stakeholders

APPENDIX: PERFORMANCE SLAs, ROLLBACK, AND ESCALATION

Performance SLA Targets

Metric	Target	Measurement Period	Remedy if Breached
System Availability	99.999% (5 min/year)	Monthly	Service credit 10% monthly fee
P1 Incident Response	<15 minutes	Per incident	Service credit 5% monthly fee
On-Site Escalation	<4 hours	Per P1 incident	Service credit 5% monthly fee
Auth Latency (ADS-B)	<50 ms end-to-end	Weekly average	Investigation and remediation
False Positive Rate	<0.001%	Monthly average	Threshold re-tuning within 5 days
Revocation Propagation	<60 seconds	Per event	Emergency process review
AI Model Accuracy	>99.7% systematic	Quarterly	Model retraining within 30 days
Software Patch Deployment	<72 hours critical	Per vulnerability	Emergency maintenance window

Rollback Procedures

Trigger	Rollback Action	Time to Rollback	Authority Required
Authentication node failure	Automatic failover to standby node	<500 ms	None (automated)
False positive rate >0.01%	Revert to advisory-only mode	<30 minutes	ANSP Ops Manager
Safety net conflict detected	Suspend authentication display	<5 minutes	ANSP Safety Manager
PKI infrastructure failure	Revert to behavioral-only mode	<60 seconds	ANSP Ops Manager
Authority instruction to suspend	Full system bypass/pass-through	<5 minutes	ANSP Accountable Mgr
Major software update failure	Rollback to previous certified version	<2 hours	QBITEl Change Board

Escalation Matrix

Priority	Definition	QBITEl Response	Customer Contact	Authority Notification
P1 — Critical	Auth system down or safety impact	<15 min / 24x7 SOC	ANSP Ops Director	Immediate if safety impact
P2 — High	Detection rate degraded or high FP rate	<1 hour / 24x7 SOC	ANSP IT Manager	Within 24 hours if required
P3 — Medium	Non-critical feature issue	<4 hours / business hours	Technical lead	Not required
P4 — Low	Enhancement or cosmetic issue	<2 business days	Account manager	Not required
Security Incident	Confirmed spoofing attack detected	<5 min / SOC escalation	ANSP Safety Manager	Per national authority requirements

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